Expressions

Letters stand for numbers.

Exponents: how many times to multiply

$$3^4 = 3 \times 3 \times 3 \times 3 = 81$$

The small 4 says: use 3 as a factor 4 times. 3^2 is “3 squared” = 9.

Write it with a letter

5 more than n	$n + 5$
7 less than n	$n - 7$
3 times n	$3n$
n split into 4 equal parts	$n \div 4$

Find the value: put the number in for the letter

When $n = 5$: $3n + 4 = 3 \times 5 + 4 = 19$

Same value, different look

$2a + 5a = 7a$

$3(x + 4) = 3x + 12$

Your turn!

When $n = 6$: $2n + 7 =$ ☐

Expressions: Practice

A letter stands for a number.

Part 1 Exponents Write it as multiplication, then find it.

1) $2^5 = 2 \times 2 \times 2 \times 2 \times 2 =$ ☐

2) $3^3 = 3 \times 3 \times 3 =$ ☐

3) $5^3 = 5 \times 5 \times 5 =$ ☐

4) $4^3 = 4 \times 4 \times 4 =$ ☐

5) $6^2 = 6 \times 6 =$ ☐

6) $2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 =$ ☐

7) $7^2 = 7 \times 7 =$ ☐

8) $3^4 = 3 \times 3 \times 3 \times 3 =$ ☐

9) $9^2 = 9 \times 9 =$ ☐

10) $8^2 = 8 \times 8 =$ ☐

11) $2^4 = 2 \times 2 \times 2 \times 2 =$ ☐

Expressions: Practice (continued)

Part 1 Exponents (continued)

12) $5^4 = 5 \times 5 \times 5 \times 5 =$ ☐

Part 2 Write it with a letter Fill in the number.

13) 6 more than n: n + ☐

14) 9 less than n: n − ☐

15) n times 8: ☐ n

16) n shared by 5: n ÷ ☐

17) 12 more than t: t + ☐

18) 4 less than y: y − ☐

19) double k: ☐ k

20) 10 times m: ☐ m

21) p split into 3 parts: p ÷ ☐

22) 15 added to r: r + ☐

Expressions: Practice (continued)

Part 3 Find the value Put the number in for the letter.

23) when n = 5: 3n + 4 = ☐

24) when n = 9: 2n + 7 = ☐

25) when n = 6: 5n + 3 = ☐

26) when n = 8: 4n + 1 = ☐

27) when n = 3: 6n + 10 = ☐

28) when n = 4: 7n + 5 = ☐

29) when n = 2: 8n + 9 = ☐

30) when n = 5: 9n + 6 = ☐

31) when n = 12: 2n + 11 = ☐

32) when n = 11: 3n + 8 = ☐

33) when n = 7: 4n + 12 = ☐

34) when n = 9: 5n + 2 = ☐

Expressions: Practice (continued)

Part 3 Find the value (continued)

35) when n = 8: 6n + 7 = ☐

36) when n = 4: 10n + 13 = ☐

Part 4 Put like terms together No help this time.

37) 2a + 5a = ☐ a

38) 4x + 3x = ☐ x

39) 6y + 7y = ☐ y

40) 3n + 9n = ☐ n

41) 8b + 4b = ☐ b

42) 5k + 6k = ☐ k

43) 7m + 8m = ☐ m

44) 9p + 3p = ☐ p

45) 10t + 5t = ☐ t

46) 11z + 4z = ☐ z

Expressions: Practice (continued)

Part 4 Put like terms together (continued)

47) $6c + 12c = \square\, c$

48) $13r + 2r = \square\, r$

Part 5 Open the parentheses Multiply each part.

49) $3(x + 4) = 3x + \square$

50) $5(y + 2) = 5y + \square$

51) $2(n + 9) = 2n + \square$

52) $6(a + 3) = 6a + \square$

53) $4(m + 7) = 4m + \square$

54) $7(b + 5) = 7b + \square$

55) $8(t + 2) = 8t + \square$

56) $9(k + 3) = 9k + \square$

57) $2(p + 11) = 2p + \square$

58) $10(r + 6) = 10r + \square$

Expressions: Practice (continued)

Part 6 More values

59) when n = 3: 12n + 16 = ☐

60) when n = 2: 9n + 19 = ☐

61) when n = 6: 4n + 8 = ☐

62) when n = 10: 11n + 12 = ☐

63) when n = 3: 11n + 19 = ☐

64) when n = 5: 10n + 13 = ☐

65) when n = 12: 11n + 9 = ☐

66) when n = 8: 7n + 5 = ☐

67) when n = 3: 12n + 17 = ☐

68) when n = 2: 6n + 10 = ☐

69) when n = 4: 8n + 4 = ☐

70) when n = 6: 6n + 1 = ☐

Expressions: Practice (continued)

Part 6 More values (continued)

71) when n = 12: 8n + 3 = ☐

72) when n = 6: 4n + 5 = ☐

73) when n = 2: 11n + 2 = ☐

74) when n = 12: 8n + 2 = ☐

75) when n = 4: 12n + 3 = ☐

76) when n = 10: 7n + 14 = ☐

Part 7 More like terms

77) 9y + 3y = ☐ y

78) 11g + 7g = ☐ g

79) 8g + 6g = ☐ g

80) 12x + 10x = ☐ x

81) 8p + 13p = ☐ p

82) 9t + 2t = ☐ t

Expressions: Practice (continued)

Part 7 More like terms (continued)

83) $4c + 7c = \square\, c$

84) $14c + 6c = \square\, c$

85) $4w + 4w = \square\, w$

86) $14k + 15k = \square\, k$

87) $12a + 13a = \square\, a$

88) $11y + 5y = \square\, y$

Part 8 More parentheses

89) $2 (c + 3) = 2c + \square$

90) $4 (y + 15) = 4y + \square$

91) $2 (d + 4) = 2d + \square$

92) $5 (a + 5) = 5a + \square$

93) $10 (q + 13) = 10q + \square$

94) $5 (p + 10) = 5p + \square$

Expressions: Practice (continued)

Part 8 More parentheses (continued)

95) $6\,(\,w + 10\,) = 6w + \square$

96) $4\,(\,y + 7\,) = 4y + \square$

97) $11\,(\,n + 14\,) = 11n + \square$

98) $11\,(\,w + 4\,) = 11w + \square$

Part 9 Remember? Problems from earlier packets.

99) $-16 \;\square\; -6$

100) $|\,-37\,| = \square$

101) GCF of 56 and 84 = $\square$

102) LCM of 12 and 15 = $\square$

103) $5.46 \div 0.6 = \square$

104) 80% of 35 = $\square$

Expressions: Practice (continued)

Part 10 Word problems Write an expression, then find its value.

105) Movie tickets cost $9 each. Write the cost of t tickets. What do 7 tickets cost?

Number sentence _____

Answer ☐

Draw or show your work

106) Mia is 4 years older than her brother, who is b years old. How old is Mia if b = 8?

Number sentence _____

Answer ☐ years

Draw or show your work

107) A pizza is cut into 8 slices. How many slices are in p pizzas? Find it for p = 6.

Number sentence _____

Answer ☐ slices

Draw or show your work

108) A square has sides of 9 cm. Its area is 9^2 square cm. What is the area?

Number sentence _____

Answer ☐ square cm

Draw or show your work

109) Challenge: A taxi costs $3 plus $2 a mile: 2m + 3. What does a 12-mile ride cost?

Number sentence _____

Answer ☐

Draw or show your work

Check-In: Expressions

Show what you know. No help this time. Score: _____ / 9

Part 1 Exponents

1) $3^5 = 3 \times 3 \times 3 \times 3 \times 3 =$ ☐

2) $10^2 = 10 \times 10 =$ ☐

Part 2 Find the value

3) when $n = 7$: $3n + 2 =$ ☐

4) when $n = 4$: $5n + 9 =$ ☐

5) when $n = 8$: $2n + 13 =$ ☐

Part 3 Put like terms together

6) $4w + 11w =$ ☐ w

7) $12e + 3e =$ ☐ e

Part 4 Open the parentheses

8) $4(x + 9) = 4x +$ ☐

9) $3(n + 12) = 3n +$ ☐